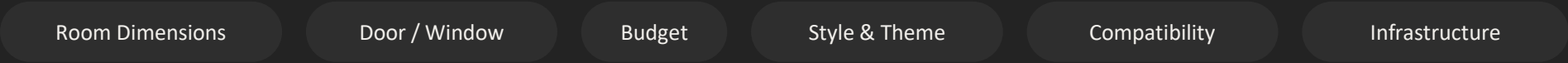


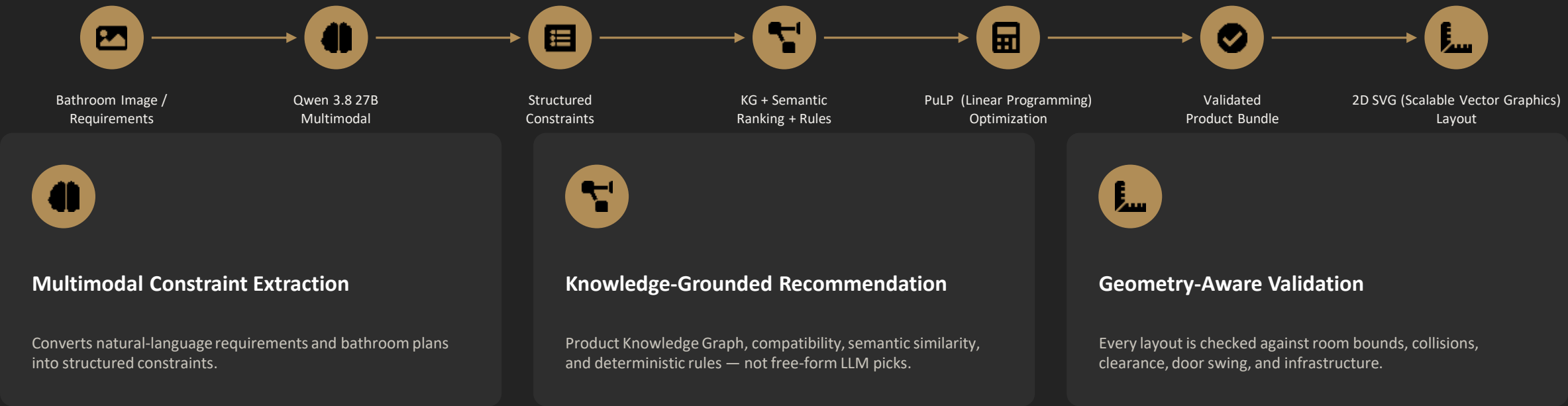
KOHLER AI BATHROOM DESIGNER AND PLANNER

An AI-powered multimodal design assistant for personalized bathroom product recommendations and spatially validated layouts

From Bathroom Intent to a Spatially Valid Product Bundle



Core Approach: Separate AI understanding from deterministic design decisions.



AI Is Used Where It Adds Intelligence — Rules Are Used Where Accuracy Matters



1 — UNDERSTAND

Qwen 3.8 27B

- Reads bathroom plans / images
- Extracts dimensions & door info
- Parses natural-language intent
- Produces structured JSON



2 — GROUND

Knowledge Graph

- Connects products, categories, styles
- Finishes, compatibility, accessories
- Infrastructure requirements (e.g. electricity, plumbing)
- Grounds recs in product knowledge



3 — DECIDE

Hybrid Recommendation

- Semantic relevance
- Explicit user constraints
- Compatibility & budget rules
- Integer optimization (PuLP)



4 — VERIFY + VISUALIZE

Deterministic Geometry

- Generate candidate layouts
- Validate hard spatial constraints
- Reject invalid layouts
- Render selected layout as SVG

Never ask the LLM to hallucinate product compatibility or geometric feasibility.

INTERACTIVE REFINEMENT WORKFLOW

Replace one product

Rerank alternatives

Re-optimize bundle

Regenerate layout

Revalidate geometry

For detailed System and Module Architecture, please refer to the GitHub README.

Tech Stack and Execution



AI / INTELLIGENCE

- Groq
- Qwen 3.8 27B
- Multimodal extraction
- Structured JSON
- Sentence Transformers
- Prompt-driven requirement parsing



ENGINEERING

- Python · Flask · Pydantic
- Neo4j AuraDB · PuLP
- Deterministic geometry engine
- SVG renderer
- Modular provider abstraction
- Graceful offline fallback path



USER EXPERIENCE

- Natural-language design request
- Upload bathroom plan / image
- Personalized product bundle
- Budget-aware alternatives
- 2D layout visualization
- Product replacement workflow

EXECUTION

- Modular architecture
- Automated test suite — 45 tests passing
- Structured validation between AI and deterministic components
- Graceful fallback when live LLM functionality is unavailable
- Designed for extension to real product catalog data

An AI design assistant that combines multimodal understanding with knowledge-grounded recommendations and deterministic spatial validation.

From Customer Pain Point to Competitive Advantage

PROBLEM

Bathroom Layout/Design planning is a constraint-satisfaction problem customers can't solve alone.

Room fit, budget, style, and installation compatibility must align simultaneously · Generic planners visualize OR filter catalogs, never both · Wrong purchases mean returns and lost trust

SOLUTION

One assistant that recommends, optimizes, and validates — together.

Multimodal understanding of the room and intent · Knowledge-grounded recommendation via KG, semantic ranking, and optimization · Deterministic geometry guarantees the result is real, not hallucinated

BUSINESS IMPACT

Higher AOV

Required-accessory bundling surfaces add-ons automatically, at the point of decision.

Fewer Returns

Compatibility and physical fit are verified before purchase, not discovered after.

Showroom-Ready

The same engine powers online self-serve and in-person dealer consultations.

Extensible

The KG + optimization + geometry pattern generalizes to kitchens and other categories.